

performance and dropped it in favour of a regular 256-bit memory bus. The 128 SPs were still present, but the new card was something of a sterilised stud with the ancient 8800 GTX still defeating it in most benchmarks. The big attraction was the price. While the 90nm G80 was still commanding a price of Rs 28,000 or so, the 9800 GTX debuted at a price of Rs 25,000, and has constantly fallen. Today, cards based on this chipset can be purchased for as low as Rs 11,000, making the 9800 GTX a perfect candidate for a good gaming setup for those with shallow pockets.

There are also some cards with a *plus* "+" as a suffix. NVIDIA chose this rather confusing name to designate the newer 55nm parts — after all it needed a scapegoat to test the functionality of the 55nm fab, before moving its costlier parts (like the GTX 2xx series) to this process — thus the 9800 GTX+ was born. Being a smaller fab, NVIDIA ramped up clock speeds keeping the same thermal envelopes — the 9800 GTX+ runs at a core speed of 738 MHz, while the 9800 GTX runs at a speed of 675 MHz.

The GeForce 9800 GT is a 55/65nm part, and features the same 112 SPs that the 8800 GT had. The sweetener to an already sugary pie is the price — this card's based on the GPU retail price tag of Rs 9,000 or thereabouts. It adds support for hybrid power saving. What this means is that the GPU can be shut down totally on a GeForce motherboard thereby saving power, and then the onboard graphics will take over. While gaming, you can switch to the GPU. Hybrid SLI is also possible — where the onboard GPU also contributes to the graphics card's processing power, although this obviously only works with combinations of NVIDIA cards and motherboards.

The 8800 GT is the old man in this collection of GPUs. It was a lower variant of the G80 core, featuring 112 SPs and a 256-bit bus. It is built around an older 80nm fabrication process.



Zotac 9800 GTX AMP! EDITION

Features

There were seven 9800 GTX cards in this comparison, out of which five were the plus versions i.e. 9800 GTX+. An oddity was that the *GPU-Z* — a utility which gives complete information about your graphics card specifications and features — showed the 9800 GTX+ cards as 65nm versions. We are not sure if there are two versions of these GPUs, but according to NVIDIA's own information the 9800 GTX+ is a 55nm part. Whether it's a 55nm only part is unclear. This aside, all the 9800 GTXs in our test had stock NVIDIA coolers, and most were heavily overclocked as well, the highest overclocked part being the XFX PV-T98W-YDQH which was clocked at 765 MHz on the core. All these brands provide a full connector bundle, except the XFX PV-T98F-YDF8 which missed out on component and S-Video connectors. Sad to say only XFX and ZOTAC bothered to provide any games — the ZOTAC 9800 GTX AMP! Edition came bundled with *Lost*, while the Plus version came with *XIIth Century Death Or Glory*. XFX provided *Company Of Heroes*, while also including a disk which contained the DX10 patch for the game — we found this really neat.

There were six 9800 GT based cards, out of which ASUS' offering was the only 55nm version according to *GPU-Z*. Gigabyte

and Palit offered 1 GB versions, while the rest made do with 512 MB. ASUS' EN 9800 GT HYBRID POWER was factory overclocked to 650 MHz (600 MHz default), while ZOTACs GeForce 9800 GT AMP! Edition was overclocked to a superb 700 MHz, giving this card at least a theoretical advantage in terms of performance. ZOTAC's offering was titled *Limited Edition*. We're told the special orange coloured PCB used is a ten-layer one, as opposed to an eight-layer design used on other cards. This ensures that this card is more stable while overclocking and probably improves the life of the card as well, contributing to its build quality. When looking at the prices of the 9800 GT cards we realise that NVIDIA is serious about causing a stir in the mid-range segment by introducing products at entry level prices. The Palit 9800 GT is priced at just Rs 8,250 which, for a 1 GB card is pretty good value. EVGA's 9800 GT has perhaps one of the best looking stickers we've seen a 9800GT, and its card stands out, even though it sports a reference NVIDIA cooler. Priced at Rs 9,600 the EVGA e-GeForce 9800 GT is also very well priced. Gigabyte's 9800GTs are a little too expensive, even more so than some of the 9800 GTX cards. ZOTAC was the only brand among the 9800 GT vendors to bundle a game — *XIIth Century Death Or Glory*. The EN8800 GT TOP sports an